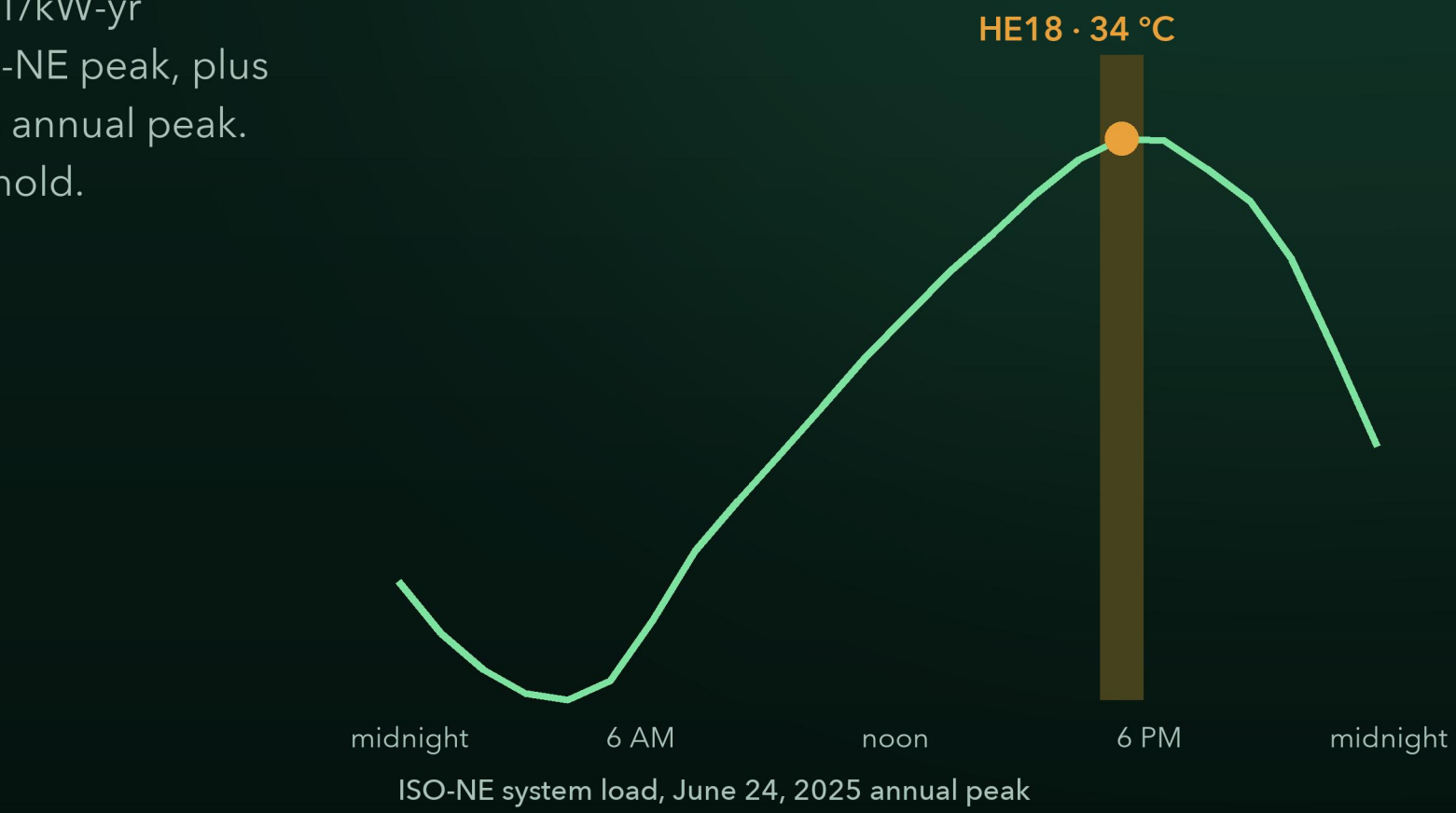
# PeakSure × REON EBS

Reliable, affordable peak power

THE PROBLEM

# One hot hour sets a town's power bill

Massachusetts utilities pay $183.71/kW-yr transmission on each monthly ISO-NE peak, plus capacity ($3.58/kW-month) on the annual peak. Those charges reach every household.




ISO-NE system load, June 24, 2025 annual peak

WHERE REON COMES IN

# A battery is only as strong as its weakest cell

In a fixed series string the first weak cell stops the whole string. REON puts a switch at every cell: a weak cell is bypassed and the rest keep delivering.

Fixed string: stops at the 70 Ah cell



REON bypass: weak cell switched out



**+20% energy with bypass: 84.06 vs 70 Ah per cell (challenge example)**

PeakSure × REON EBS · simulation on real ISO-NE data

OUR SOLUTION · TWO PHASES

# Prove it on today's batteries, then build it into tomorrow's

## Phase 1 · Retrofit

Add REON's cell-level BMS to batteries already installed.

- Prove the technology works
- Cheapest way in: no new battery

evidence

## Phase 2 · Scale

Sell REON EBS batteries with cell control built in.

- Once the field evidence is in
- PeakSure as the decision layer

PHASE 1 · RETROFIT

# Add cell-level control to batteries that already exist

## Cost

About $25–75 per kWh (team estimate).

No new battery to buy.

Exit test: more energy delivered and fewer missed peaks, measured on site.

### Where it pays (model, per year)

| | |
|---|---|
| 1-h battery | +$24,000 |
| Failing cooling | +$98,000 |
| Faulty cells | +$168,000 |
| Healthy 2-h battery | +$7,000 |

### Massachusetts candidates

Sterling: 2 MW / 3.9 MWh (2016)

Reading: 5 MW / 10 MWh (2019)

Holden: 5 MW / 22 MWh (2024)

PHASE 2 · SCALE

# With proof in hand, sell REON EBS with cell control built in

## When

Phase 1 shows measured extra energy and fewer missed peaks.

REON confirms EBS specifications and certification.

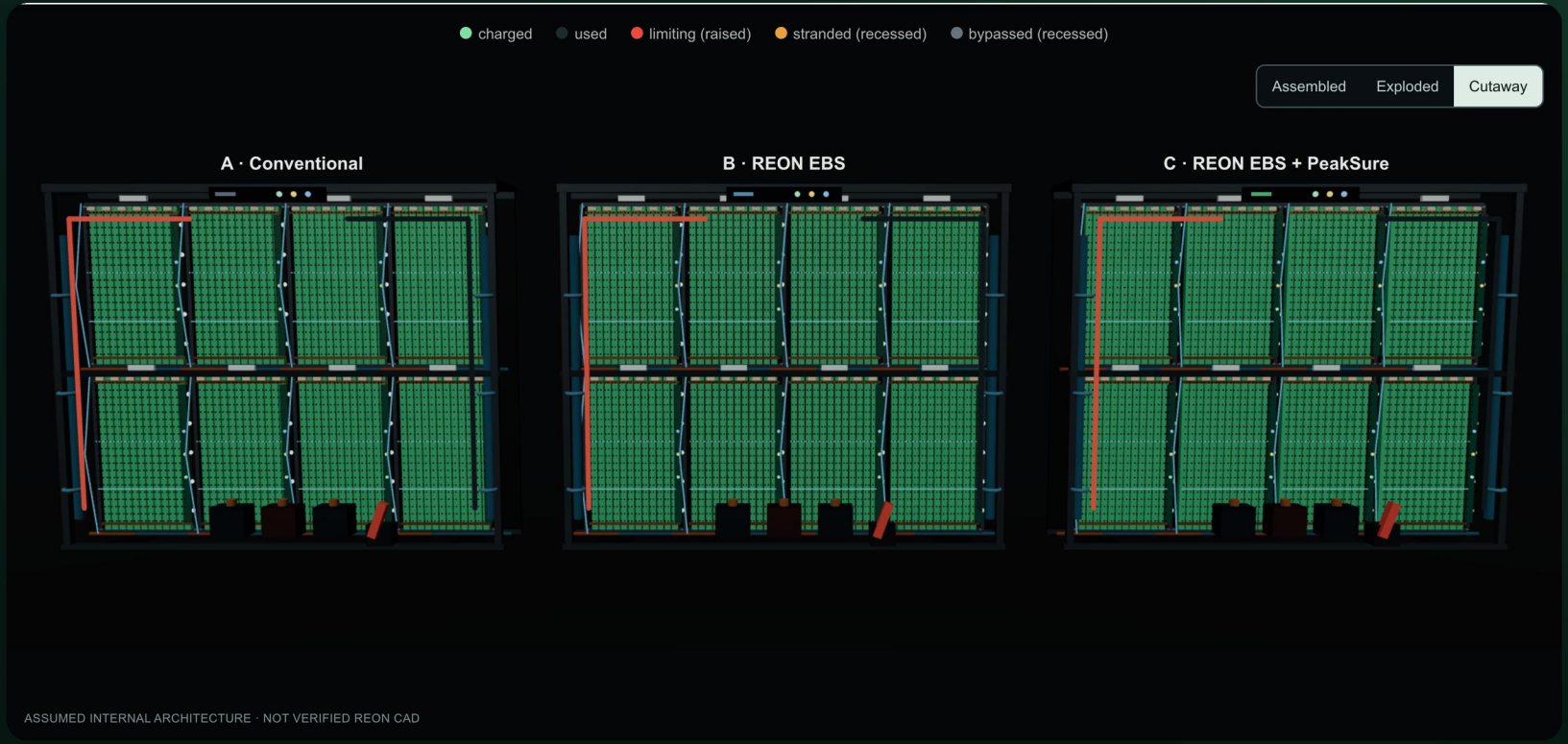
## Where

- New municipal light plant storage in Massachusetts
- Commercial sites on demand charges, e.g. National Grid G-2 at $15.24/kW-month
- Priced -5% to +20% vs conventional (team estimate)

COMPARE TAB · LIVE MODEL

# The real June 24, 2025 peak hour

HE18 (17:00–18:00) · 34 °C · same cells in all three batteries




**A · Conventional**
Delivered 958 kWh
Strings on 8/8

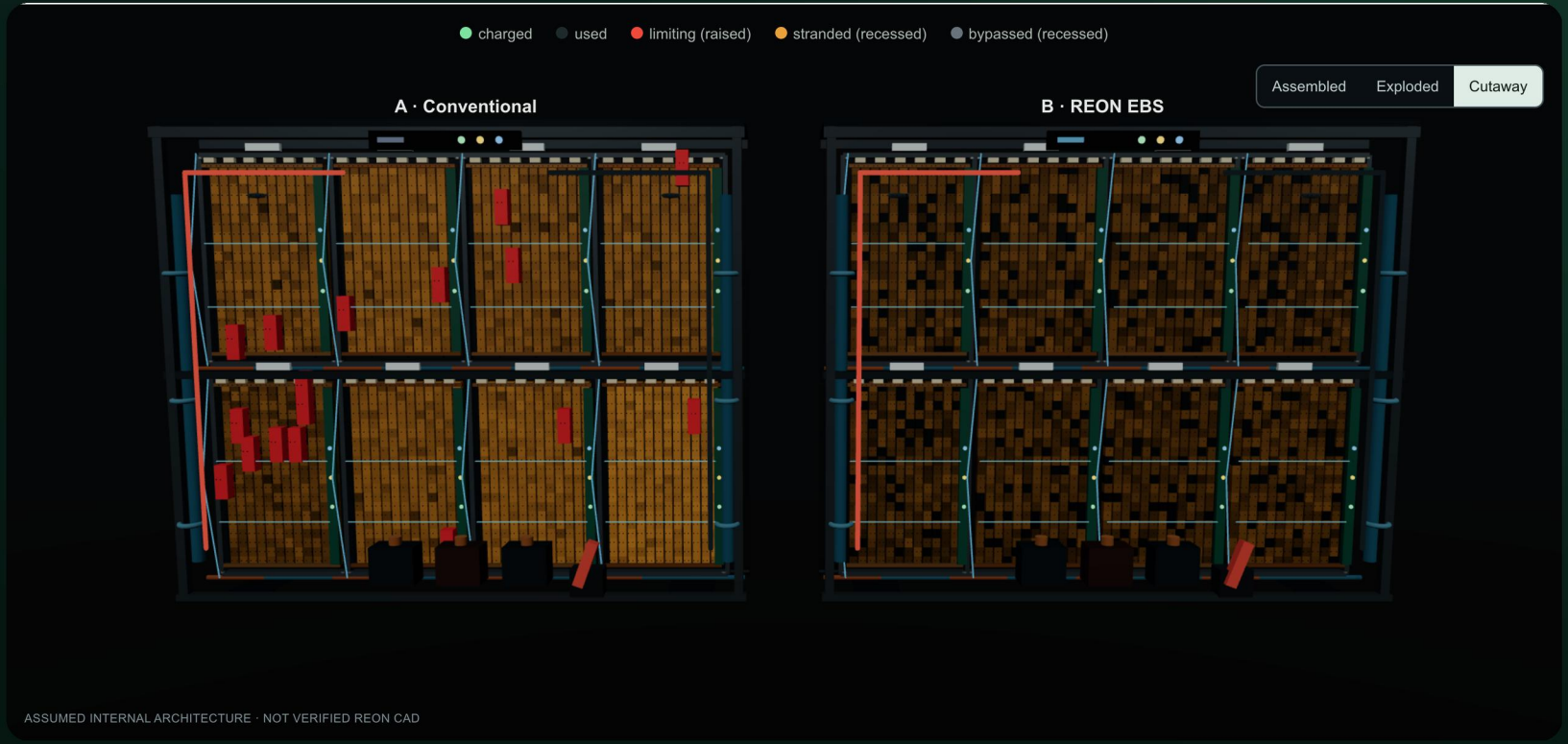
**B · REON EBS**
Delivered 958 kWh
Strings on 8/8

**C · REON EBS + PeakSure**
Delivered 958 kWh
Strings on 8/8

COMPARE TAB · LIVE MODEL

# Drain test: where the energy gets stranded

Full power until every string stops · conventional vs REON EBS · red = the cell that stopped its string




**A · Conventional**

Delivered 3,855 kWh

Stranded 638 kWh

**B · REON EBS**

Delivered 4,221 kWh

Stranded 201 kWh

ASSUMPTIONS TAB

# Every assumption, labelled

BY CASE

| | A · Conventional | B · REON EBS (bypass) | C · REON EBS + PeakSure | |
|---|---|---|---|---|
| Wiring | Fixed series strings: first empty cell stops its string | Per-cell bypass: string runs while ≥ N_min cells | Same as B | ASSUMED |
| What the operator knows | Annual whole-battery capacity test | Annual whole-battery capacity test | Per-cell twin (±2% cell estimate) | ASSUMED |
| How it commits | Full power for as many hours as P5 allows | Same as A | Best expected saving with 95% delivery confidence | ASSUMED |
| Cells | Identical cells in all three cases (same seed and draws) | | | ASSUMED |
| Layout (simulation abstraction) | 8 strings × 240 cells | 8 × 240, N_min 216 (10% headroom) | Same as B | ASSUMED |
| Switch loss | None | 1.0% (≈ I·R_on / V_cell) | Same as B | ASSUMED |
| Inverter efficiency | 96% | 96% | 96% | ASSUMED |
| Cooling | Container 23 °C setpoint, k = 0.25; cell spread σ 2.5 °C | Same | Same | ASSUMED |

OUR RECOMMENDATION

# Start now with a retrofit pilot on batteries that are already struggling

**NEXT 3 MONTHS**

Read-only data pilot

**3–12 MONTHS**

Retrofit pilot on 1–2 MA batteries

**AFTER THE EVIDENCE**

Scale with REON EBS

| | |
|---|---|
| **Reliability** | covers 73% of monthly peaks vs 66% |
| **Affordable power** | more from batteries towns already own |
| **Scaling** | every pilot turns modelled value into measured value |

# Prove it on today's batteries.

Build it into tomorrow's.